

The augmented edge signature

A wandering 3×3 family for $\Phi = J^{-1} + U^T(A^\dagger)^{-1}U$, not a fixed matrix limit

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Abstract

The mixed form of round 369 writes $M^\dagger = R^\dagger - \frac{1}{2}I = A^\dagger + UJU^T$ with an explicit 3×3 signature $J = \text{diag}(1, 1, 1/\text{den})$ and U the last two dual Christoffel–Darboux columns together with the Sherman–Morrison border. Haynsworth (proved in `RH/Haynsworth.lean`) converts positivity of $M^\dagger$ into a comparison of $\text{ind}_- \Phi$ with $\text{ind}_- A^\dagger$, $\Phi = J^{-1} + U^T(A^\dagger)^{-1}U$. The 3×3 Φ is not a restatement of that identity: after an explicit Sylvester map D_k on the K_2 block and a scale $d_3 = 1/\sqrt{|\text{sch}|}$ it equals a two-parameter family $\Phi_{\text{edge}}(a, b)$ whose inertia is constant on each chart, whose spectral gap to 0 is uniformly positive on any bounded set of (a, b) , and whose reconstruction error is identically zero. On the named census the couplings (a, b) stay in a compact set with $g^* \geq 0.08$; dead χ leave the living chart by $\text{sch} > 0$. No RH claim.

Status. Lemma 1 and Lemma 2 are proved (finite 3×3 algebra). Proposition 1 is a finite machine check, not a cofinal law. No RH claim. No L^* claim. No $R^\dagger \succ \frac{1}{2}I$ claim.

1 The 3×3 object

Write $A_0 = R_{N-3} - \frac{1}{2}I$ on the hole nodes Y , U_{CD} the last two dual CD columns, and $A^\dagger = \text{blkdiag}(A_0, -\frac{1}{2})$. Sherman–Morrison on the bordered dual resolvent supplies a third column $w = (s; -1)$ and $\text{den} = 1 + \gamma - v^T R s$, with $J = \text{diag}(1, 1, 1/\text{den})$. Then [1]

$$\Phi = \begin{pmatrix} K_2 & c_0 \\ c_0^T & \varphi_{bb} \end{pmatrix}, \quad K_2 = I + U_{\text{CD}}^T A_0^{-1} U_{\text{CD}},$$

and $K_2 = (I + R_+) - ww^T/\lambda_-$ on the $\text{nneg}(A_0) = 1$ branch, with $\det K_2 = \det(I + R_+)(1 - \gamma/\lambda_-)$ [2]. The Schur complement of K_2 in Φ is the scalar

$$\text{sch} = \varphi_{bb} - c_0^T K_2^{-1} c_0.$$

Haynsworth on Φ itself reads $\text{ind}_- \Phi = \text{ind}_- K_2 + \text{ind}_-(\text{sch})$. On the named construction $\text{ind}_- K_2 = \text{ind}_- A_0$, so

$$\text{ind}_- \Phi = 1 + \text{ind}_- A_0 \iff \text{sch} < 0. \quad (1)$$

The $+1$ is the border mode. It is not the macroscopic $\frac{1}{2}$ -cluster of R_{N-3} (round 398): that cluster sits in A_0 from above and is bulk.

2 Explicit D and the model family

Let $K_2 = V \text{diag}(\mu) V^T$ be the 2×2 spectral theorem and $D_2 = V \text{diag}(1/\sqrt{|\mu|})$. Set $d_3 = 1/\sqrt{|\text{sch}|}$ on $\text{sch} \neq 0$ and $D = \text{blkdiag}(D_2, d_3)$. Then $D^T K_2 D$ in the leading 2×2 is $J_2 = \text{diag}(\text{sign } \mu)$, and the off-block of $D^T \Phi D$ is a pair $(a, b) \in \mathbb{R}^2$.

Lemma 1 (reconstruction). *Let J_2 be diagonal with entries ± 1 , let $\tau = (a, b)^T$, and let $s = \text{sign}(\text{sch})$. The unique symmetric 3×3 with leading block J_2 , off-block τ and Schur complement s is*

$$\Phi_{\text{edge}}(\tau) = \begin{pmatrix} J_2 & \tau \\ \tau^T & s + \tau^T J_2^{-1} \tau \end{pmatrix}. \quad (2)$$

Hence $D^T \Phi D = \Phi_{\text{edge}}(a, b)$ identically (the error operator is zero).

This is not a fit and not a fixed matrix M_0 . The family wanders with $\tau = (a, b)$.

Lemma 2 (model lemma). *On the P1 chart $J_2 = \text{diag}(-1, 1)$, $s = -1$,*

$$\text{ind}_- \Phi_{\text{edge}}(a, b) = 2 = 1 + \text{ind}_- A_{\text{edge}}$$

for every $(a, b) \in \mathbb{R}^2$, where $A_{\text{edge}} = (-1)$ records the single negative of K_2 . On the vacuous chart $J_2 = I_2$, $s = -1$, $\text{ind}_- \Phi_{\text{edge}} = 1 = 1 + 0$. In both charts $\det \Phi_{\text{edge}} = \det(J_2) s = \pm 1 \neq 0$.

Proof. Haynsworth additivity on (2): $\text{ind}_- \Phi_{\text{edge}} = \text{ind}_- J_2 + \text{ind}_-(s)$. The P1 chart contributes $(1) + (1)$; the vacuous chart $(0) + (1)$. The determinant identity is the Schur formula. On the rational point $(a, b) = (0, 0)$ one has $\Phi_{\text{edge}} = \text{diag}(-1, 1, -1)$ with inertia $(1, 2, 0)$, and $\Phi_{\text{edge}}(1, 0)$ has determinant 1. $\square$

The gap $g(\tau) = \text{dist}(0, \sigma(\Phi_{\text{edge}}))$ is positive on any bounded $K \subset \mathbb{R}^2$. Along the a -axis, $g(a, 0) = (2 + a^2 - |a|\sqrt{a^2 + 4})/2 \sim 1/a^2$. A 64-point check on the circle of radius 3.2 gives $g^* \geq 0.082$ and inertia 2 at every sample.

3 The coordinate τ

Three candidates were tested, in the order asked.

(i) *The r375 pair.* $\rho = \gamma/\lambda_-$ and $\delta = \det(I + R_+)$, projectively $\xi = (\rho - 1)/(\rho + 1)$, $\eta = \delta/(1 + \delta)$. On the $\text{nneg} = 1$ branch of core-42, $\xi \in [0.121, 0.622]$, $\eta \in [0.768, 0.991]$, $\text{den} \in [1.46, 1.63]$, halves-medians trend-free. An isotropic reconstruction of Φ from (ρ, δ) alone has $\|E\|/g = 80.7$ at FRAME-A: the pair does not determine the 3×3 . Spearman of $|(a, b)|$ against ξ is 0.12.

(ii) *The r321 sliding coordinate.* A QMAX proxy of the CD columns has Spearman 0.26 against $|(a, b)|$ and Spearman -0.96 against N (it shrinks). It is not the 3×3 modulus.

(iii) *The two signature couplings.* $\tau = (a, b)$ as in §2. On core-42 (P1), $|(a, b)| \leq 3.144$; the disk of radius 3.2 is a compact K with $g^* \geq 0.08$. Large- N EXT windows have *smaller* $|(a, b)|$ (0.075 at $N = 1119$, 0.203 at $N = 3022$). The set K is stable in N (r360-conform: τ may wander inside a fixed compact).

4 Census and two-sidedness

Proposition 1 (named census). *On core-42, a six-row EXT sample, Twin $kz \in \{9, 18, 52\}$, living χ_3 (37) and living χ_4 (41): $J \succ 0$, Haynsworth balance, (1), and $\|E\| \leq 10^{-10}$. Dead χ_3 (5/5) and dead χ_4 -20 have $\text{sch} > 0$ and violate (1). Scramble has $\text{ind}_- A_0 = 21$ and the border pack dies at step 37. Two-period $S = 21$ has $\text{ind}_- A_0 = 4$. Dropping the border column yields shift 0 already on the rational 4-node model.*

The reconstruction error is a SATZ (Lemma 1), not a shrinking remainder. Cofinality of K is not proved.

5 Named kills

Dead χ : $\text{sch} > 0$, so the border mode of Φ is positive — the shift fails, or $\text{ind}_- \Phi = 0$ (dead χ_4 -20). Several dead rows have $|(a, b)|$ up to 6.7, outside living K . Scramble: $\text{ind}_- A_0 = 21$ leaves the edge chart $\{0, 1\}$; the SM pack is not positive-prefix. Two-period: $\text{ind}_- A_0 = 4$. Mutants: omitting the border column kills the $+1$; the wrong-sign J_{33} is already caught in Fractions by round 369 (residual 867/115).

Machine check

Standalone: Fractions mixed form, mixed Haynsworth, both charts of Φ_{edge} at the origin, determinant 1, drop-border, 64-point disk. Construction: FRAME-A shift and pins, reconstruction $\|E\| = 0$, dead χ_3 -15, scramble 21, two-period, living χ_3 -9. Discovery census: `edge_signature_probe.py` (`--smoke` / `full`). Exit line: `EDGE SIGNATURE VERIFIED`.

Claim boundary

Finite identities for the mixed form, for Haynsworth, for the 3×3 reconstruction and for the model lemma on both charts; a named census that the living worlds land in a compact K with $g^* > 0$ and that dead χ leave by $\text{sch} > 0$. No statement about zeros of $\zeta(s)$, in either direction. No RH claim.

References

- [1] S. Hamann, *Mixed Haynsworth*, sealed probe, August 2026, `experiments/tfpt-discovery/mixed_haynsworth_probe.py`.
- [2] S. Hamann, *The P2 lemma*, `rh/problem/p2_lemma_proof.tex`.
- [3] `RH/Haynsworth.lean`, mixed inertia additivity, proved.